Chapter 9 — Planar Graphs

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1 Plane and Planar Graphs

1.1 Abstract Graphs vs. Embedded Graphs

A graph $G = (V, E)$ is an abstract combinatorial object consisting of a vertex set V and an edge set E. By itself, a graph contains no geometric, metric, or topological information. In particular, there is no intrinsic notion of:

- distance,
- angle,
- faces or regions,
- or spatial arrangement.

Topological structure arises only after a *drawing* or *embedding* of the graph in the plane is specified.

1.2 Planar Graphs

A graph G is called *planar* if it admits at least one drawing in the plane in which edges intersect only at common endpoints. Formally,

$$\exists \text{ an embedding of } G \text{ in } \mathbb{R}^2 \text{ with no edge crossings.}$$

Planarity is therefore an *existential property.* It depends solely on the abstract adjacency structure of G and not on any particular drawing.

Remark 1.1. A graph may be drawn with crossings and still be planar, provided that a crossing-free drawing exists.

1.3 Plane Graphs

A *plane graph* is a specific embedding of a planar graph in the plane with no crossings.
Thus:

- **Planar** refers to a property of an abstract graph.
- **Plane** refers to a particular embedding of that graph.

Every plane graph is planar, but a planar graph may admit multiple non-equivalent plane embeddings.

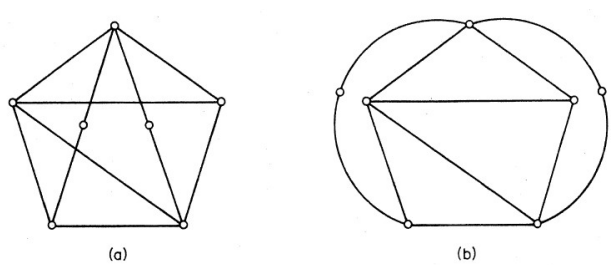

Figure 9.1. (*a*) A planar graph G; (*b*) a planar embedding of G

1.4 Multiple Embeddings

Two embeddings of the same planar graph need not be equivalent. For example, the cycle C_4 can be embedded:

- as a square,
- or in a distorted configuration that changes the arrangement of faces.

Although the underlying graph is identical, the cyclic order of edges around vertices and the induced face structure may differ. Such embeddings are said to be *non-equivalent*. This distinction is fundamental because many structural properties depend on the chosen embedding.

1.5 Drawings, Embeddings, and Equivalence

A drawing of a graph assigns:

- each vertex to a distinct point in the plane,
- each edge to a continuous curve joining its endpoints.

Two drawings represent the same embedding if one can be continuously deformed into the other without:

- introducing edge crossings,
- breaking or subdividing edges,
- moving vertices through edges.

This equivalence corresponds to *ambient isotopy* in topology.

1.6 Faces and Regions

Once a plane embedding is fixed, the graph partitions the plane into connected regions called *faces*. Formally, a face is a connected component of

$$\mathbb{R}^2 \setminus G.$$

Properties:

- Exactly one face is unbounded (the *outer face*).
- All other faces are bounded (inner faces).
- Faces depend on the embedding, not solely on the abstract graph.

The boundary of a face is a closed walk in G. It need not be a simple cycle and may repeat vertices or edges, particularly in the presence of bridges or cut-vertices.

1.7 Embedding-Dependent and Graph Invariants

Certain quantities depend only on the abstract graph:

- number of vertices $|V|$,
- number of edges $|E|$,
- degree sequence.

Other quantities depend on the chosen embedding:

- number of faces F,
- lengths of faces,
- the dual graph.

For a connected plane graph, Euler's Formula states:

$$|V| - |E| + F = 2.$$

While the formula holds for every plane embedding, the value of F depends on the embedding chosen.

1.8 Rotation Systems

A *rotation system* encodes a plane embedding combinatorially. For each vertex v, it specifies the cyclic order of edges incident to v as they appear around that vertex in the plane. Key facts:

- Different cyclic orders correspond to different embeddings.
- A full rotation system determines a plane embedding uniquely (up to reflection).
- Planarity testing algorithms construct rotation systems rather than geometric drawings.

Rotation systems provide a purely combinatorial description of embeddings and are central in algorithmic and topological graph theory.

1.9 Special Cases and Structural Rigidity

Planar graphs may contain:

- multiple edges,
- loops,
- bridges.

These features influence face structure but do not violate planarity. A fundamental rigidity result is **Whitney's Theorem**:

Theorem 1.1 (Whitney)**.** A 3-connected planar graph has a unique embedding on the sphere, up to reflection.

This explains why 3-connected planar graphs — in particular polyhedral graphs — possess strong structural rigidity and why duality behaves especially well in this setting.

2 Dual Graphs

2.1 Why Duality Appears Only After Plane Graphs

The notion of a dual graph cannot be defined for an abstract planar graph. It requires:

- a fixed *plane embedding*,
- a well-defined set of faces,
- a clear notion of adjacency between faces.

Duality is therefore an *embedding-dependent construction*. This is why the concept arises only after plane graphs and face structure have been established.

2.2 Definition of the Dual Graph

Let G be a connected plane graph. The *dual graph* G^* is constructed as follows:

- For each face f of G, introduce a vertex f^* in G^*.
- For each edge e of G that separates two (not necessarily distinct) faces f_1 and f_2, introduce an edge e^* in G^* joining f_1^* and f_2^*.

If an edge of G borders the same face on both sides, then the corresponding edge in G^* is a loop. Consequently,

$$|V(G^*)| = F(G), \qquad |E(G^*)| = |E(G)|.$$

2.3 Dual of the Dual

If G is a connected plane graph, then

$$(G^*)^* \cong G.$$

However, this statement must be interpreted carefully:

- The isomorphism holds only up to graph isomorphism.
- The same embedding must be used.
- The statement fails for disconnected graphs unless treated componentwise.

Thus duality is *involutive*, but not canonical at the level of abstract planar graphs.

2.4 Bridges and Loops: A Fundamental Correspondence

Duality exchanges bridges and loops.

- If e is a bridge in G, then both sides of e belong to the same face. Hence e^* is a loop in G^*.
- If e is a loop in G, then e creates a new face bounded by itself. The corresponding dual edge e^* is therefore a bridge in G^*.

This correspondence plays a central role in structural graph theory.

2.5 Degree Correspondence

Let f be a face of G. Then

$$\deg_{G^*}(f^*) = \text{length of the boundary walk of } f.$$

Important:

- The boundary may repeat vertices.
- The boundary may repeat edges.
- It is a closed walk, not necessarily a simple cycle.

Thus face structure in G translates directly into vertex degrees in G^*.

2.6 Embedding Dependence

The same planar graph may admit non-equivalent embeddings whose dual graphs are non-isomorphic. For example, K_4 admits embeddings in which:

- all faces are triangular,
- or one face is a quadrilateral.

The corresponding dual graphs differ structurally. Therefore:

There is no "the" dual of a planar graph. There is only the dual of a *plane* graph.

2.7 Spherical Perspective

Embedding graphs on the sphere rather than the plane removes the asymmetry of the outer face. On the sphere:

- all faces are treated symmetrically,
- dual of dual coincides with the original embedding (up to reflection),
- each dual edge intersects exactly one primal edge.

This is the standard viewpoint in topological graph theory.

2.8 Geometric Interaction of Primal and Dual

In a plane embedding:

- Each dual edge e^* intersects exactly one primal edge e.
- No two dual edges intersect.
- The union of G and G^* forms a cellular decomposition of the plane (or sphere).

This geometric picture is fundamental in:

- planar separator theory,
- electrical network theory,
- topological graph theory.

3 Bridges in Plane Graphs

Bridges were previously introduced as edges whose removal increases the number of connected components. In planar graph theory, however, bridges acquire additional structural significance.

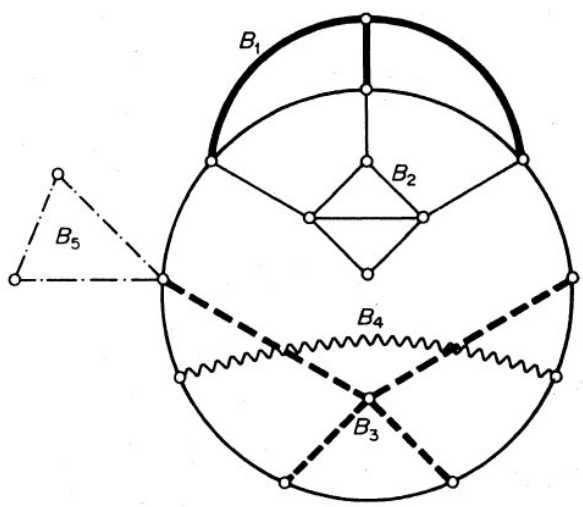

Figure 1: Bridges in a Graph

3.1 Why Bridges Reappear

In a plane graph, bridges influence:

- the structure of face boundaries,
- the appearance of loops in the dual graph,
- embedding rigidity,
- decomposition into blocks.

Thus bridges are not merely connectivity defects; they are structurally visible in the embedding.

3.2 A Fundamental Planar Property

Let G be a connected plane graph and let e be a bridge of G. Then:

- The same face lies on both sides of e.
- The boundary walk of that face traverses e twice (once in each direction).

Idea. Since e is a bridge, removing it disconnects the graph. Hence it cannot lie on a cycle. Therefore e does not separate two distinct faces, and both sides of e must belong to the same face. □

Consequences.

- Face boundaries must be defined as *closed walks*, not cycles.
- In the dual graph, the edge e^* corresponding to a bridge is a loop.

This explains the dual correspondence:

$$\text{bridge in } G \quad \longleftrightarrow \quad \text{loop in } G^*.$$

3.3 Blocks and Embedding Rigidity

A *block* is a maximal 2-connected subgraph.

- Every connected graph decomposes into blocks joined at cut-vertices.
- Bridges are precisely the edges not contained in any block.
- The block structure forms a tree (the block–cut tree).

3.4 Bridges and Face Counting

Although Euler's formula remains valid in the presence of bridges, face-length arguments require care.
Since bridges contribute twice to a single face boundary:

- naive inequalities based on face lengths may fail,
- structural arguments typically isolate or contract bridges first.

This is why many extremal results assume 2-connectedness.

4 Kuratowski's Theorem

Kuratowski's theorem provides the fundamental structural characterization of planar graphs.

4.1 Statement

Theorem 4.1 (Kuratowski)**.** A finite graph G is planar if and only if it contains no subgraph that is a subdivision of:

$$K_5 \quad \text{or} \quad K_{3,3}.$$

Such subdivisions are called *Kuratowski subgraphs*.

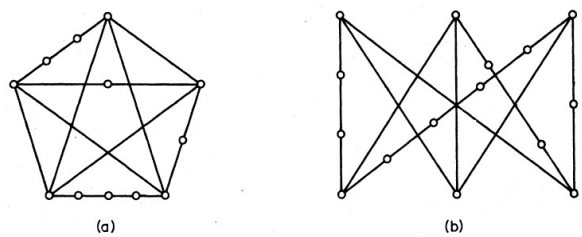

Figure 9.16. (*a*) A subdivision of K_5; (*b*) a subdivision of $K_{3,3}$

- K_5 is the minimal nonplanar complete graph.
- $K_{3,3}$ is the minimal nonplanar bipartite graph.

They represent two fundamentally different obstructions:

- **density obstruction** (K_5),
- **bipartite crossing obstruction** ($K_{3,3}$).

Every nonplanar graph must contain one of these configurations in disguised form.

4.2 Subdivisions

A *subdivision* of a graph is obtained by replacing edges with internally vertex-disjoint paths. Equivalently:

- Edges may be subdivided into chains of degree-2 vertices.
- The essential adjacency pattern is preserved.

Subdivisions preserve nonplanarity. Thus planarity is fundamentally a *topological* property.

4.3 Role of Cycles and Bridges in the Proof

A classical proof strategy proceeds as follows:

1. Assume G is a minimal nonplanar graph.
2. Choose a longest cycle C in G.
3. Analyze the bridges of C (subgraphs attached to C).
4. Show that certain configurations of bridges (especially skew bridges) force a subdivision of K_5 or $K_{3,3}$.

Thus the earlier analysis of bridges relative to a cycle is not incidental — it is structurally central to the proof.

4.4 Conceptual Importance

Kuratowski's theorem:

- characterizes planarity purely combinatorially,
- explains why Euler-type inequalities detect nonplanarity,
- underlies modern planarity testing algorithms,
- reveals that nonplanarity arises from specific forbidden configurations.

5 The Five-Colour Theorem

5.1 Statement

Theorem 5.1 (Five-Colour Theorem)**.** Every planar graph is vertex-colourable with at most five colours.

Formally, for every planar graph G, there exists a colouring function

$$c : V(G) \to \{1, 2, 3, 4, 5\}$$

such that

$$c(u) \neq c(v) \quad \text{whenever } uv \in E(G).$$

This result is weaker than the Four-Colour Theorem, but its proof is elementary and entirely combinatorial. This result is weaker than the Four-Colour Theorem, but its proof is elementary and entirely combinatorial.

5.2 Historical Context

The theorem was proved by Percy John Heawood in 1890. Its proof relies on:

- Euler's formula,
- the existence of low-degree vertices in planar graphs,
- Kempe chains.

The key structural input is:

$$E \leq 3V - 6 \quad \Longrightarrow \quad \text{average degree} < 6 \quad \Longrightarrow \quad \exists v \text{ with } \deg(v) \leq 5.$$

Thus every planar graph contains a vertex of degree at most five.

5.3 Proof by Induction

We prove the theorem by induction on $|V(G)|$.

Base Case. Graphs with $|V| \leq 5$ are trivially 5-colourable.

Inductive Hypothesis. Assume every planar graph with $n - 1$ vertices is 5-colourable.

Inductive Step. Let G be a planar graph with n vertices.
By the degree argument, choose a vertex v with

$$\deg(v) \leq 5.$$

Remove v to obtain a planar graph G'. By induction, G' admits a 5-colouring. We now reinsert v.

Case 1: $\deg(v) \leq 4$

At most four colours appear among the neighbours of v. One colour remains unused, and we assign it to v.

Case 2: $\deg(v) = 5$

Suppose the five neighbours of v are coloured with five distinct colours

$$1, 2, 3, 4, 5.$$

Since G is planar, the neighbours of v appear in a cyclic order around v in the embedding. We apply the *Kempe chain argument*.

Consider the neighbours coloured 1 and 3 (which are nonconsecutive in the cyclic order). Let $H_{1,3}$ be the subgraph induced by vertices coloured 1 or 3. If the two neighbours of v coloured 1 and 3 lie in different connected components of $H_{1,3}$, then:

- Swap colours 1 and 3 in the component containing the neighbour coloured 1.
- Colour v with colour 1.

The colouring remains proper.

If instead they lie in the same component, then a Kempe chain connects them. Because of planarity and the cyclic order of neighbours, this prevents certain other pairs from being connected. In that case, we instead consider colours 2 and 4 and repeat the argument. Planarity ensures that not both alternating pairs can form connecting Kempe chains; otherwise, crossings would be forced in the embedding. Hence at least one colour swap succeeds, and v can be coloured.

By induction, every planar graph is 5-colourable.

6 Kempe Chains

Let $G = (V, E)$ be a graph with a proper colouring c.
Let a, b be two colours.

The (a, b)-*Kempe chain* containing a vertex v (with $c(v) = a$) is the connected component of v in the subgraph induced by vertices coloured only a or b. Equivalently, it consists of all vertices reachable from v via paths whose colours alternate between a and b.

- **Maximality**: The Kempe chain is maximal with respect to connectivity using only colours a and b.
- **Colour Swapping**: If we interchange colours a and b on all vertices of a Kempe chain, the colouring remains proper. Every edge inside the chain joins vertices of different colours. Edges leaving the chain connect to vertices of colours different from a and b, so no conflict is created.
- **Locality**: Two vertices of colour a may lie in different (a, b)-Kempe chains if no alternating path connects them.

6.1 Role in the Colour Theorems

The difficulty arises when a vertex v has five neighbours coloured with five distinct colours.
Kempe chains allow a local rearrangement of colours:

- Choose two non-adjacent colours in the cyclic order around v.
- Attempt a Kempe swap.
- If one swap is blocked by connectivity, planarity guarantees another pair will succeed.

Thus Kempe chains provide a mechanism for freeing a colour without destroying properness.

Graph colouring is not rigid — colours can be globally reorganised through local connectivity.

In the Five-Colour Theorem, this flexibility compensates for the maximal degree 5 constraint imposed by Euler's formula.

Kempe originally attempted to prove the Four-Colour Theorem using this method. The argument contained a subtle flaw for the four-colour case. Originally posed as a conjecture in 1852 by Francis Guthrie, the Four-Colour Theorem was eventually proved in 1976 (Appel–Haken) using extensive computer-assisted case analysis. The Five-Colour Theorem remains the clean, purely theoretical precursor.

7 The Four-Colour Theorem

Theorem 7.1 (Four-Colour Theorem)**.** Every planar graph is vertex-colourable with at most four colours.

Formally, for every planar graph G, there exists a colouring function

$$c : V(G) \to \{1, 2, 3, 4\}$$

such that

$$c(u) \neq c(v) \quad \text{whenever } uv \in E(G).$$

Early Attempts.

- Kempe (1879) proposed a proof using what are now called Kempe chains. The argument contained a subtle flaw.
- Heawood (1890) corrected the argument sufficiently to prove the Five-Colour Theorem.

Although Kempe's proof failed for four colours, his method became central to later developments.

The 1976 Proof. Kenneth Appel and Wolfgang Haken established the theorem using computer-assisted verification. Their strategy:

- Reduce the problem to a finite collection of configurations.
- Prove that each configuration is reducible.
- Verify reducibility via computer.

The original proof required checking 1,936 configurations.

7.1 Reduction to a Minimal Counterexample

The proof proceeds by *contradiction*. Assume there exists a planar graph that is not 4-colourable. Let G be a minimal counterexample (minimal in number of vertices). Then:

- G is connected.
- G is simple.
- Every proper subgraph of G is 4-colourable.

By Euler's formula and the inequality

$$E \leq 3V - 6,$$

G must contain a vertex of degree at most 5.
Hence every minimal counterexample must contain vertices of degree 5. The structural analysis focuses on how degree-5 vertices can be arranged.

7.2 Unavoidable Sets and Reducible Configurations

Two central notions are introduced:

Unavoidable Set. A collection of configurations is unavoidable if every planar graph must contain at least one of them.

Reducible Configuration. A configuration is reducible if it cannot appear in a minimal counterexample; that is, assuming its presence allows one to extend a 4-colouring from a smaller graph.

The strategy is:

1. Exhibit an unavoidable finite set of configurations.
2. Prove each configuration in the set is reducible.

If both statements hold, no minimal counterexample can exist.

7.3 The Discharging Method

The discharging method is the central structural tool.

Initial Charge Assignment. Assign to each vertex v a charge

$$\mathrm{ch}(v) = \deg(v) - 6.$$

By summing over all vertices and using Euler's formula,

$$\sum_{v \in V} (\deg(v) - 6) = -12.$$

Thus the total charge is negative.

Redistribution (Discharging Rules). Charges are redistributed locally according to carefully designed rules, preserving total charge. The goal is to show that, after redistribution, every vertex has nonnegative charge — contradicting the total sum being negative. This contradiction implies that a forbidden configuration must exist.

8 Non-Hamiltonian Planar Graphs

Definition 8.1 (Planar Graph)**.** A graph G is *planar* if it admits an embedding in the plane with no two edges crossing except at common endpoints.

Definition 8.2 (Hamiltonian Cycle)**.** Let $G = (V, E)$ be a finite simple graph with $|V| = n$. A *Hamiltonian cycle* in G is a cycle

$$v_1, v_2, \ldots, v_n, v_1$$

such that:

1. $\{v_i, v_{i+1}\} \in E$ for all $i = 1, \ldots, n-1$,
2. $\{v_n, v_1\} \in E$,
3. $v_i \neq v_j$ for all $i \neq j$.

Thus every vertex of G appears exactly once in the cycle.

Definition 8.3 (Non-Hamiltonian Graph)**.** A graph is *non-Hamiltonian* if it does not contain a Hamiltonian cycle.

Definition 8.4 (Non-Hamiltonian Planar Graph)**.** A graph that is both planar and non-Hamiltonian.

8.1 Key Results

Whitney's Theorem (1931)

Every 4-connected planar graph is Hamiltonian.
Thus sufficiently strong connectivity *does* force Hamiltonicity in planar graphs.

Tutte's Counterexample (1956)

Theorem 8.1 (Tutte)**.** There exists a 3-connected planar graph that is non-Hamiltonian.

Tutte constructed a 3-regular (cubic), 3-connected planar graph with 46 vertices that contains no Hamiltonian cycle. This disproved the conjecture that 3-connectivity suffices for Hamiltonicity in planar graphs.
Later constructions reduced the minimum number of vertices required.

Grinberg's Theorem (1968)

Grinberg's theorem gives a necessary condition for Hamiltonicity in planar graphs.

Theorem 8.2 (Grinberg)**.** Let G be a planar graph with a Hamiltonian cycle C. For $k \geq 3$, let:

- f_k = number of faces of length k inside C,
- g_k = number of faces of length k outside C.

Then

$$\sum_{k \geq 3} (k-2)(f_k - g_k) = 0.$$

If a planar graph violates this identity, it cannot be Hamiltonian. *Grinberg's theorem provides a powerful obstruction tool for constructing non-Hamiltonian planar graphs.*

8.2 Structural Observations

Connectivity.

- 2-connectivity is necessary for Hamiltonicity.
- 3-connectivity is not sufficient (Tutte).
- 4-connectivity *is* sufficient (Whitney).

Thus Hamiltonicity in planar graphs lies at a delicate threshold of connectivity.

Degree Conditions. Unlike Dirac's theorem for general graphs, no simple minimum-degree condition (below full density) guarantees Hamiltonicity in planar graphs. Even cubic planar graphs (all degrees 3) may be non-Hamiltonian.

Face Structure. The combinatorics of face lengths strongly influence Hamiltonicity. Certain distributions of face sizes force violation of Grinberg's identity, preventing Hamiltonian cycles. Small faces (triangles, quadrilaterals) arranged improperly can create structural bottlenecks that block global traversal.

8.3 Construction Techniques

Tutte Fragments

Tutte's construction replaces edges or vertices with carefully engineered subgraphs ("Tutte fragments") that preserve **planarity** and **3-connectivity** but destroy Hamiltonicity by forcing unavoidable parity or traversal constraints.

Face-Based Constructions

Using Grinberg's theorem:

1. Choose a planar embedding.
2. Prescribe face lengths strategically.
3. Ensure that
$$\sum_{k\geq 3}(k-2)(f_k - g_k) \neq 0.$$

This guarantees that no Hamiltonian cycle can exist.

9 Grinberg's Theorem

Let G be a connected planar graph with a fixed planar embedding.
Assume G contains a Hamiltonian cycle C.
Since C is a simple cycle passing through every vertex exactly once, it is a Jordan curve in the plane. Therefore, it separates the plane into:

- an *interior* region,
- an *exterior* region.

Every face of the embedding lies either inside or outside C, except for the two regions bounded directly by C itself.

Notations For $k \geq 3$, define:

- f_k = number of faces of length k inside C,
- g_k = number of faces of length k outside C.

Here, the *length* of a face means the number of edges on its boundary.

- Triangle $\Rightarrow k = 3$
- Quadrilateral $\Rightarrow k = 4$
- Pentagon $\Rightarrow k = 5$

9.1 Statement of Grinberg's Theorem

Theorem 9.1 (Grinberg, 1968)**.** If a planar graph G contains a Hamiltonian cycle C, then

$$\sum_{k \geq 3} (k-2)(f_k - g_k) = 0.$$

Interpretation. Each face contributes a weight $(k-2)$ depending on its size. The theorem asserts that the weighted difference between interior and exterior faces must vanish.

9.2 Important Remarks

- The condition is **necessary**, but not sufficient.
- If the sum is nonzero, no Hamiltonian cycle exists.
- If the sum equals zero, Hamiltonicity is still not guaranteed.

(1) Embedding matters. Grinberg's condition depends on the chosen planar embedding. Different embeddings may yield different face counts.

(2) Large graphs. The theorem is most effective for:

- cubic planar graphs,
- graphs with many small faces,
- highly structured constructions (e.g., Tutte-type graphs).

(3) Why $(k-2)$? The term $(k-2)$ reflects deviation from triangulation. A triangle contributes weight 1, a quadrilateral weight 2, etc. The identity balances total "excess boundary length" inside and outside C.

Grinberg's theorem converts a global cyclic property (Hamiltonicity) into a purely combinatorial constraint on face sizes.
It is therefore:

- a structural obstruction theorem,
- derived from Euler-type counting,
- one of the most effective tools for constructing non-Hamiltonian planar graphs.